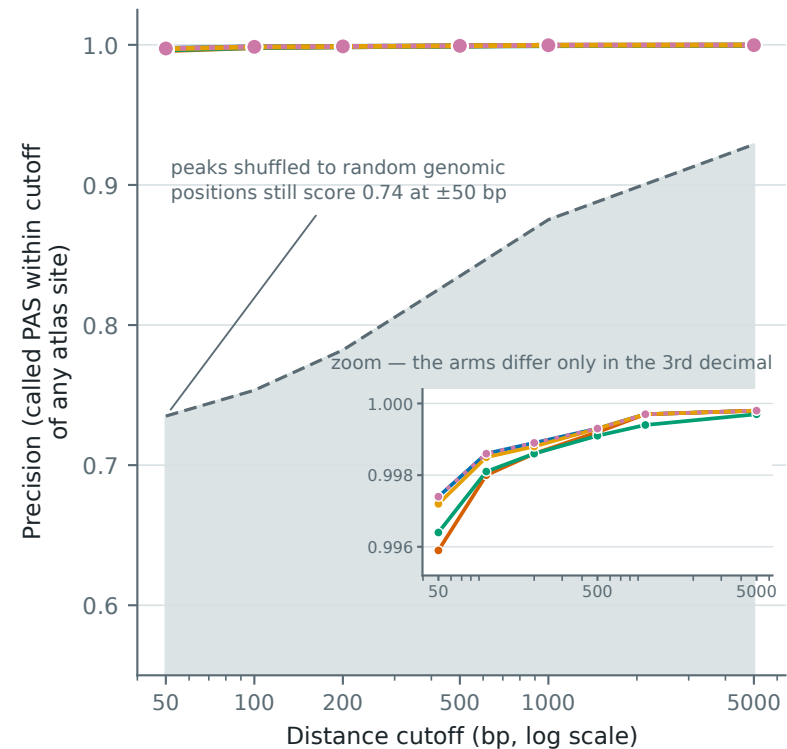
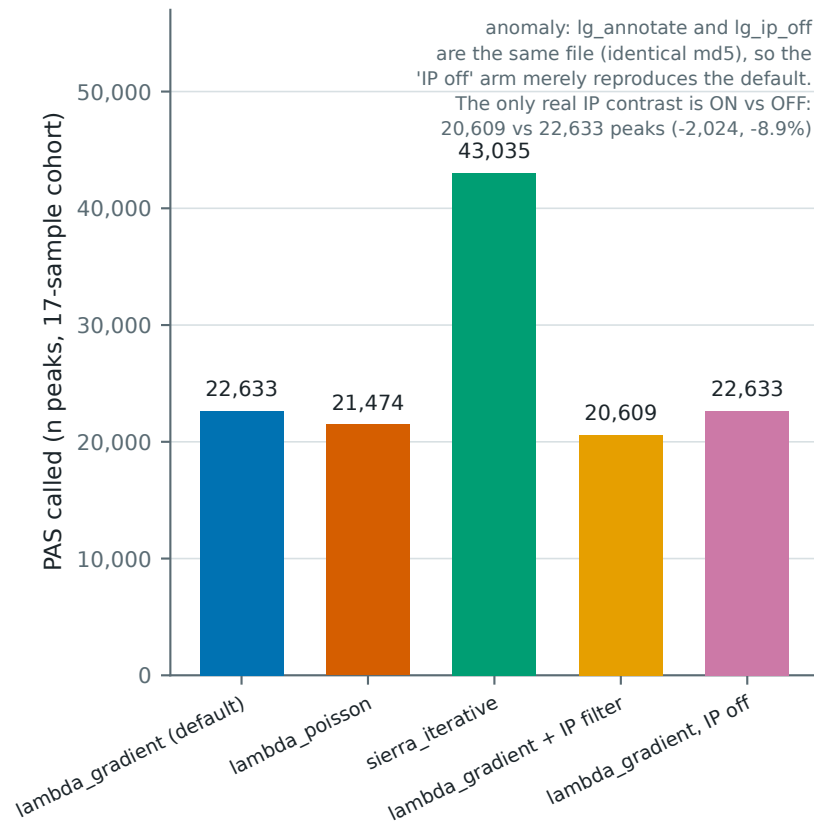


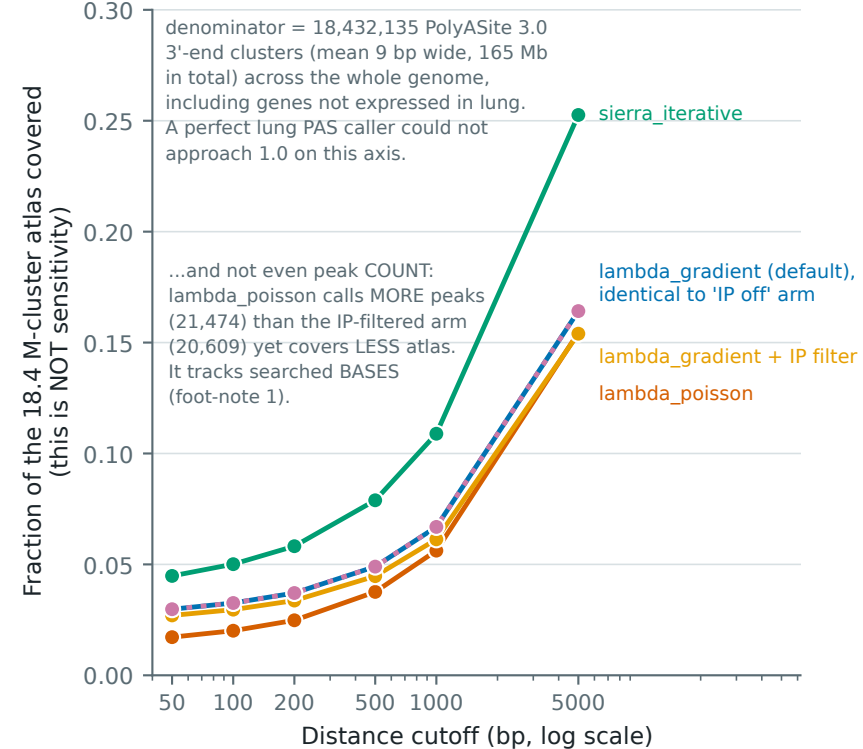
a Every strategy saturates precision — but so does randomly shuffled noise



b sierra_iterative calls 1.90–2.00× as many PAS as the lambda methods; the IP filter removes 8.9%



c 'Recall' here tracks how much sequence each arm searches — it is not a sensitivity estimate



—●— lambda_gradient (default) —●— lambda_poisson —●— sierra_iterative —●— lambda_gradient + IP filter —●— lambda_gradient, IP off — chance floor: same peaks shuffled genome-wide (3 seeds)

What this benchmark cannot show | (1) The reference is 18,432,135 PolyASite 3.0 3'-end clusters (mean 9 bp wide, 165 Mb), genome-wide, so panel c is atlas coverage, not sensitivity: across arms it tracks the bases searched — rank corr. 1.00 at every cutoff vs 0.80–1.00 for peak count. (2) Precision is inflated by reference density — atlas ± 50 bp windows already tile 38.6% of the genome (92.7% at ± 5 kb) — and called peaks are 403–640 bp wide (median), so the effective match window is peak width + $2 \times$ cutoff, not $2 \times$ cutoff. (3) Precision of ~ 0.999 must therefore be read against the 0.74 chance floor in panel a; it is NOT a clean accuracy claim. (4) Matching is strand-agnostic (bedtools window, no -s). (5) One cohort-level run per arm: no replicates, no CI on the arm curves.